

General Purpose Input/Output (GPIO)

EMBEDDED — Lezione 6

General-purpose and peripheral pin functions

- Pin multiplexing selects which internal function reaches the package pin.



Fig. 1: Function select: GPIO / UART TX / ADC / ... / SPI verso Pin.

GPIO mode: software reads Typical devices: LEDs, switches, keypads, simple control signals.

Peripheral mode: the pin connects to on-chip hardware such as a timer, UART, SPI, I²C, or ADC input; the peripheral controls the signals and timing.

GPIO (General Purpose I/O)

- A **GPIO port** connects memory-mapped registers to configurable electrical circuitry at the package pin.

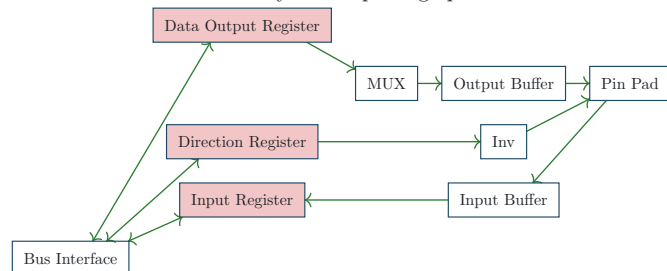


Fig. 2: GPIO Controller: bus interface, registers, MUX, output/input buffer, pin pad.

GPIO Direction and Data Registers

- Minimum of 3 registers associated with each GPIO port: **Data IN, Data Out, and Direction**.
 - The **Direction** (PxDIR) register is used to make the pin either input or output.
 - Data registers to actually write to the pin (PxOUT) or read data from the pin (PxIN).

MSP GPIO Ports

- The GPIO ports in MSP432 are designated as port **P1** to **P10** and **PJ**.
 - P1 to P10 are also referred as the **Simple I/O** or **Digital I/O** ports.
 - Port J has special function such as external crystal oscillator and JTAG connections.
- Each port contains **eight numbered pins**.

Commonly Used Registers of Port1

- The **base address** of I/O port is 0x4000 4C00.

Address	Name	Description	Type	Reset Value
0x4000 4C00	P1IN	Port1 Input Register	R	0b00000000
0x4000 4C02	P1OUT	Port1 Output Register	R/W	0b00000000
0x4000 4C04	P1DIR	Port1 Direction Register	R/W	0b00000000
0x4000 4C06	P1REN	Port1 Resistor Enable Register	R/W	0b00000000
0x4000 4C08	P1DS	Port1 Drive Strength Register	R/W	0b00000000
0x4000 4C0A	P1SEL0	Port1 Select 0 Register	R/W	0b00000000
0x4000 4C0C	P1SEL1	Port1 Select 1 Register	R/W	0b00000000

Register field descriptions

Field	Bits	Type	Reset	Description
PxDIR	7-0	RW	0h	Port X direction. 0b = Port configured as input; 1b = Port configured as output.
PxOUT	7-0	RW	Undefined	Port X output. Output mode: 0b = Output is low, 1b = Output is high. Input mode with pullups/pulldowns enabled: 0b = Pulldown selected, 1b = Pullup selected.
PxREN	7-0	RW	0h	Port X pullup or pulldown resistor enable. When the port is configured as an input, setting this bit enables or disables the pullup or pulldown. 0b = Pullup or pulldown disabled; 1b = Pullup or pulldown enabled.
PxSEL0	7-0	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of the PxSEL1 and PxSEL0 bit positions are combined to specify the function. See PxSEL1 for the definition of each value.
PxSEL1	7-0	RW	0h	Port function selection. 00b = General-purpose I/O is selected; 01b = Primary module function is selected; 10b = Secondary module function is selected; 11b = Tertiary module function is selected.
PxIN	7-0	R	Undefined	Port X input. 0b = Input is low; 1b = Input is high.

Bit-mask Operations

- Use **read-modify-write** operations when other pins in the same port must retain their configuration.

Operation	Expression	Effect
Set selected bits	REG = MASK	Selected bits become 1
Clear selected bits	REG &= ~MASK	Selected bits become 0
Toggle selected bits	REG ^= MASK	Selected bits invert
Read selected bits	REG & MASK	Nonzero when any selected bit is 1
Replace whole register	REG = value	Every bit receives a new value

GPIO as Input

MSP432 LAUNCHPAD

- Buttons are connected to **P1.1** AND **P1.4**.

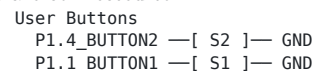


Fig. 3: User Buttons: P1.4_BUTTON2 (S2) e P1.1_BUTTON1 (S1) verso GND.

Configure User Button (P1.1)

- Configure P1.1 as input port
 - BIT1 = 0000 0001
 - P1->DIR = ~BIT1;
- Select Pullup mode.
 - P1->OUT = BIT1;
- Enable Pullup Resistor
 - P1->REN = BIT1;
- Set GPIO Mode
 - P1->SEL0 = 0;
 - P1->SEL1 = 0;
- Check Input Register to detect the input

```

/* Configure P1.1 as input*/
P1->DIR = ~BIT1;
/* Select pullup*/
P1->OUT = BIT1;
/* Enable pullup*/
P1->REN = BIT1;
/* Set as I/O */
P1->SEL0 = 0;
P1->SEL1 = 0;
...
/* Catch Button Press */
while (P1->IN & BIT1);
while (!(P1->IN & BIT1));
...

```

Pullup and Pulldown Resistors

- If we do not use internal **pullup** (or **pulldown**) resistors, we have a problem.
 - We need to ensure a known value on the output if a pin is left **floating**.
- We want the switch SW to pull the pin to ground, so we enable the pull-up.
 - The pin value is:
 - High when SW is not pressed
 - Low when SW is pressed

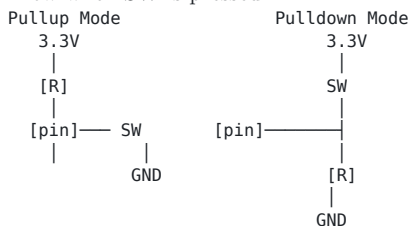


Fig. 4: Pullup Mode e Pulldown Mode: pin, resistore [R], switch SW, GND, 3.3V.

Alternate functions

- Pins may have **alternate functions**: UART, SPI, and I2C, timers...
- SEL1 and SEL0 choose
 - The function **multiplexer** selects GPIO or alternate function

SEL1	SEL0	Selected function
0	0	Simple digital I/O
0	1	Primary module function
1	0	Secondary module function
1	1	Tertiary module function

Pin	Other functionality
P1.2 / P1.3	UART receive and transmit, I ² C signals
P2 pins	Timer capture or compare outputs
P4 pins	Clock, timer, and peripheral bus signals
P5 / P6 pins	Analog inputs, references, and communication signals
P7 to P10	Timer, clock, communication, and analog functions

Fig. 5: Function multiplexer: GPIO / Primary / Secondary / Tertiary Module Function, selezionati da PxSel1/PxSel0.

Lab 1

- Put breakpoints and observe the execution.

```

int main(void) {
    P1->SEL1 &= ~2;          /* configure P1.1 as simple I/O */
    P1->SEL0 &= ~2;
    P1->DIR &= ~2;           /* P1.1 set as input */
    P1->REN |= 2;            /* P1.1 pull resistor enabled */
    P1->OUT |= 2;            /* Pull up/down is selected by P1-
>OUT */
    while (1) {
        while(P1->IN & 2);
        while (!(P1->IN & 2));
    }
}

```

Bouncing GPIO Input

- Mechanical **bounce** creates several digital transitions.



Fig. 6: Bounce: BUTTON, RAW INPUT, DEBOUNCED EVENT; finestre di bounce a PRESS e RELEASE.

How to handle in software?

Strategy	How it works
Delay after edge	Wait, then sample again
Stable sample count	Accept change after N identical samples

Lab 2

- Write the code that toggles Green LED when S1 switch is pushed.

User Buttons: P1.1_BUTTON1 (S1), P1.4_BUTTON2 (S2) → GND
 User LEDs:
 P1.0_LED1 —[470, R7]—[LED1]— GND
 P2.0_RGBLED_RED —[110R, R2]—
 P2.1_RGBLED_GREEN —[16R, R3]—[LED2, EVERLIGHT_19-337]
 P2.2_RGBLED_BLUE —[24R, R4]—

Fig. 7: Pulsanti S1/S2 e catene LED con resistori.

Lab 3

- Write the code that toggles Green LED when S1 switch is pushed and toggles Blue LED when S2 is pushed.

Could you detect switch actions simultaneously?

MSP432 Port Reference

Commonly Used Registers of Port2

Address	Name	Description	Type	Reset Value
0x4000 4C01	P2IN	Port2 Input Register	R	0b00000000
0x4000 4C03	P2OUT	Port2 Output Register	R/W	0b00000000
0x4000 4C05	P2DIR	Port2 Direction Register	R/W	0b00000000
0x4000 4C07	P2REN	Port2 Resistor Enable Register	R/W	0b00000000
0x4000 4C09	P2DS	Port2 Drive Strength Register	R/W	0b00000000
0x4000 4C0B	P2SEL0	Port2 Select 0 Register	R/W	0b00000000
0x4000 4C0D	P2SEL1	Port2 Select 1 Register	R/W	0b00000000

Commonly Used Registers of Port3

Address	Name	Description	Type	Reset Value
0x4000 4C20	P3IN	Port3 Input Register	R	0b00000000
0x4000 4C22	P3OUT	Port3 Output Register	R/W	0b00000000
0x4000 4C24	P3DIR	Port3 Direction Register	R/W	0b00000000
0x4000 4C26	P3REN	Port3 Resistor Enable Register	R/W	0b00000000
0x4000 4C28	P3DS	Port3 Drive Strength Register	R/W	0b00000000
0x4000 4C2A	P3SEL0	Port3 Select 0 Register	R/W	0b00000000
0x4000 4C2C	P3SEL1	Port3 Select 1 Register	R/W	0b00000000

Commonly Used Registers of Port4

Address	Name	Description	Type	Reset Value
0x4000 4C21	P4IN	Port4 Input Register	R	0b00000000
0x4000 4C23	P4OUT	Port4 Output Register	R/W	0b00000000
0x4000 4C25	P4DIR	Port4 Direction Register	R/W	0b00000000
0x4000 4C27	P4REN	Port4 Resistor Enable Register	R/W	0b00000000
0x4000 4C29	P4DS	Port4 Drive Strength Register	R/W	0b00000000
0x4000 4C2B	P4SEL0	Port4 Select 0 Register	R/W	0b00000000
0x4000 4C2D	P4SEL1	Port4 Select 1 Register	R/W	0b00000000

Alternate Pin Functions

PxSEL1.y	PxSEL0.y	Meaning
0	0	Alternative 0 (Default) Simple I/O
0	1	Alternative 1 (UART, SPI, I ² C, ...)
1	0	Alternative 2 (Timers, ...)
1	1	Alternative 3 (ADC, Comparator, ...)

Port 1 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P1.0	Simple I/O	UCA0STE	–	–
P1.1	Simple I/O	UCA0CLK	–	–
P1.2	Simple I/O	UCA0RXD/ UCA0SOMI	–	–
P1.3	Simple I/O	UCA0TXD/ UCA0SIMO	–	–
P1.4	Simple I/O	UCB0STE	–	–
P1.5	Simple I/O	UCB0CLK	–	–
P1.6	Simple I/O	UCB0SIMO/ UCB0SDA	–	–
P1.7	Simple I/O	UCB0SOMI/ UCB0SCL	–	–

Port 2 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P2.0	Simple I/O	PM_UCA1STE	–	–
P2.1	Simple I/O	PM_UCA1CLK	–	–
P2.2	Simple I/O	PM_UCA1RXD/ PM_UCA1SOMI	–	–
P2.3	Simple I/O	PM_UCA1TXD/ PM_UCA1SIMO	–	–
P2.4	Simple I/O	PM_TA0.1	–	–
P2.5	Simple I/O	PM_TA0.2	–	–
P2.6	Simple I/O	PM_TA0.3	–	–
P2.7	Simple I/O	PM_TA0.4	–	–

Port 3 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P3.0	Simple I/O	PM_UCA2STE	–	–
P3.1	Simple I/O	PM_UCA2CLK	–	–
P3.2	Simple I/O	PM_UCA2RXD/ PM_UCA2SOMI	–	–
P3.3	Simple I/O	PM_UCA2TXD/ PM_UCA2SIMO	–	–
P3.4	Simple I/O	PM_UCB2STE	–	–
P3.5	Simple I/O	PM_UCB2CLK	–	–
P3.6	Simple I/O	PM_UCB2SIMO/ PM_UCB2SDA	–	–
P3.7	Simple I/O	PM_UCB2SOMI/ PM_UCB2SCL	–	–

Port 4 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P4.0	Simple I/O	–	–	A13
P4.1	Simple I/O	–	–	A12
P4.2	Simple I/O	ACLK	TA2CLK	A11
P4.3	Simple I/O	MCLK	RTCCLK	A10
P4.4	Simple I/O	HSMCLK	SVMHOUT	A9
P4.5	Simple I/O	–	–	A8
P4.6	Simple I/O	–	–	A7
P4.7	Simple I/O	–	–	A6

Port 5 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P5.0	Simple I/O	–	–	A5
P5.1	Simple I/O	–	–	A4
P5.2	Simple I/O	–	–	A3
P5.3	Simple I/O	–	–	A2
P5.4	Simple I/O	–	–	A1
P5.5	Simple I/O	–	–	A0
P5.6	Simple I/O	TA2.1	–	VREF+/ VeREF+/ C1.7
P5.7	Simple I/O	TA2.2	–	VREF-/Ve- REF-/C1.6

Port 6 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P6.0	Simple I/O	–	–	A15
P6.1	Simple I/O	–	–	A14
P6.2	Simple I/O	UCB1STE	–	C1.5
P6.3	Simple I/O	UCB1CLK	–	C1.4
P6.4	Simple I/O	UCB1SIMO/ UCB1SDA	–	C1.3
P6.5	Simple I/O	UCB1SOMI/ UCB1SCL	–	C1.2
P6.6	Simple I/O	TA2.3	UCB3SIMO/ UCB3SDA	C1.1
P6.7	Simple I/O	TA2.4	UCB3SOMI/ UCB3SCL	C1.0

Port 7 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P7.0	Simple I/O	PM_SMCLK/ PM_DMAE0	–	–
P7.1	Simple I/O	PM_C0OUT/ PM_TA0CLK	–	–
P7.2	Simple I/O	PM_C1OUT/ PM_TA1CLK	–	–
P7.3	Simple I/O	PM_TA0.0	–	–
P7.4	Simple I/O	PM_TA1.4	–	C0.5
P7.5	Simple I/O	PM_TA1.3	–	C0.4
P7.6	Simple I/O	PM_TA1.2	–	C0.3
P7.7	Simple I/O	PM_TA1.1	–	C0.2

Port 8 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P8.0	–	UCB3STE	TA1.0	C0.1
P8.1	–	UCB3CLK	TA2.0	C0.0
P8.2	–	–	TA3.2	C0.0
P8.3	–	–	TA3CLK	A22
P8.4	–	–	–	A21
P8.5	–	–	–	A20
P8.6	–	–	–	A19
P8.7	–	–	–	A18

Port 9 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P9.0	–	–	–	A17
P9.1	–	–	–	A16
P9.2	–	TA3.3	–	–
P9.3	–	TA3.4	–	–
P9.4	–	UCA3STE	–	–
P9.5	–	UCA3CLK	–	–
P9.6	–	UCA3RXD/ UCA3SOMI	–	–
P9.7	–	UCA3TXD/ UCA3SIMO	–	–

Port 10 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P10.0	–	UCB3STE	–	–
P10.1	–	UCB3CLK	–	–
P10.2	–	UCB3SIMO/ UCB3SDA	–	–
P10.3	–	UCB3SOMI/ UCB3SCL	–	–
P10.4	–	TA3.0	–	C0.7
P10.5	–	TA3.1	–	C0.6